MTH1003

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

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Mathematical Modelling

Module Leader: Prof John Thuburn,
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Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No Word Count specified

SECTION A

1. (a) A particle of mass $m = 1$ moves subject to the time-dependent force

$$\mathbf{F}(t) = \mathrm{e}^{-t}\,\mathbf{i} + t^2\,\mathbf{j} - \mathbf{k}.$$

At time $t = 0$ the particle has position $\mathbf{r}(0) = \mathbf{0}$ and velocity $\mathbf{v}(0) = -2\,\mathbf{i}$. Find the velocity $\mathbf{v}(t)$ and position $\mathbf{r}(t)$. Also find all of the times at which the velocity is perpendicular to the vector $\mathbf{j} + \mathbf{k}$. (8)

(b) A light elastic string has natural length l_0 and elastic modulus λ_0. State Hooke's law for this elastic string.

A light compound string AC is made up of two segments: AB with natural length l and elastic modulus 3λ; and BC with natural length $2l$ and elastic modulus λ. Compute the tension in the compound string when A and C are fixed a distance $5l$ apart. (8)

(c) A damped oscillator is governed by

$$\ddot{x} + 2k\dot{x} + 4x = 0$$

and has initial condition $x = 0$, $\dot{x} = 2$ at $t = 0$. Find the solution $x(t)$ for the two cases (i) $k = 5/2$, (ii) $k = 2$. In each case state what type of damping occurs. (14)

(d) Using suitable Taylor series, show that the finite difference derivative approximation

$$f'(x) \approx \frac{1}{6h}\left\{8f(x) - 9f(x-h) + f(x-3h)\right\}$$

is second-order accurate. (7)

(e) A projectile is launched with initial speed V at an angle α to the horizontal. Starting from Newton's second law, taking g to be the gravitational acceleration, and neglecting air resistance, derive the trajectory equation

$$z = x\left(\tan\alpha - \frac{gx}{2V^2}(1+\tan^2\alpha)\right),$$

where x and z are the horizontal and vertical coordinates in the plane of motion.

When $V = \sqrt{10gH}$ and the ground is horizontal the projectile lands a distance $5H$ from the point of launch. Find the possible launch angles α.

A wall is now constructed at $x = 4H$. How tall must the wall be to prevent the projectile from reaching its previous landing point $(x, z) = (5H, 0)$, assuming the same launch speed? (13)

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SECTION B

2. (a) A particle of mass m, position $x(t)$, moves in the x-direction under the influence of a force $F(x)$. Starting from Newton's second law, derive the energy equation

$$\frac{1}{2}mv^2 + V(x) = E = \text{const},$$

where $v = \dot{x}$, and define $V(x)$ in terms of $F(x)$. (6)

(b) The force $F(x)$ is given by

$$F(x) = \frac{a}{x^3} - \frac{b}{x},$$

for constants a and b.

(i) Find the potential $V(x)$ corresponding to this force. The particle has speed U at $x = 1$ and also has speed U at $x = 2$. Show that there is an equilibrium position at

$$x = x_{\text{eq}} = \sqrt{\frac{8}{3}\ln 2},$$

where ln is the natural logarithm. (7)

(ii) The same particle also has speed $U/2$ at $x = 1/2$. Express a and b in terms of U. (4)

(iii) Hence determine the frequency at which a particle would make small-amplitude oscillations about x_{eq}. (4)

(iv) Define the work done by the force on the particle. Evaluate the work done by the above force $F(x)$, in terms of U, as the particle moves from $x = 1/2$ to $x = 1$. (4)

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3. (a) Using Lagrange's interpolation formula, or otherwise, find the quadratic function $y(x)$ that interpolates the three data points $(x_1, y_1) = (0, 12)$, $(x_2, y_2) = (1, 3)$, $(x_3, y_3) = (2, -2)$. (6)

(b) Consider the Fixed Point Iteration method for the solution of the equation $x = g(x)$:

$$x^{(m+1)} = g(x^{(m)}),$$

where m is the iteration number.

(i) Show that a necessary condition for convergence of the fixed point iteration to a root x^* is $|g'(x^*)| < 1$. (9)

(ii) Now suppose we wish to solve

$$x = \frac{1}{\ln x}$$

by taking $g(x) = 1/\ln x$. Show that the solution must satisfy $x^* > 1$. Hence deduce that the Fixed Point Iteration method will not converge for this problem. (7)

(iii) Show that the problem of part (ii) may be rewritten as

$$x = \exp(1/x),$$

and that the necessary condition for convergence of Fixed Point Iteration is satisfied when the problem is written in this way. (3)

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4. Consider the following dynamical system:

$$\dot{x} = (2 - x)(1 - y^2); \qquad \dot{y} = (1 - x)(2 + y).$$

(a) Sketch the nullclines of the system and hence find all the equilibrium points. (6)

(b) Compute the Jacobian matrix of the system and hence determine the stability type of each equilibrium point. (6)

(c) Find a conserved quantity for this dynamical system. (5)

(d) State briefly what is meant by the 'stable manifold' and 'unstable manifold' of a saddle point. Show that the system has a saddle point whose stable and unstable manifolds satisfy the equation

$$|x - 2||2 + y|^3 \exp(x - 2y + y^2/2) = 27 \exp(-1/2).$$

(8)

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